A checked classifier for support tickets

Albert's Marketplace | 10-minute defence

What we built, what the local evidence says, and what remains unknown

01 | BUSINESS PROBLEM

The task is routing, not answering

Albert's Marketplace support teams receive free-text tickets that need to reach the right queue quickly. The component predicts a controlled route such as billing, delivery, returns and refunds, account, seller or product information. Support operations and agents use the result to triage work; it does not make a policy promise to the customer.

Input	A support ticket written in free text
Output	Label, confidence and human-review flag
Success	The predicted queue matches the checked expected label

The business value is faster triage with an explicit human-review path when confidence is low or the label is other.

A classifier fits the information in the ticket

We compared classification with structured extraction and retrieval. Classification is the smallest component that matches the task: the routing label is already implied by the ticket, while the component does not need to answer from a changing policy or catalogue.

Choice	Why it fits	Consequence
Prompt-driven classifier	Maps ticket text to a fixed set of queues	Needs labelled cases and a clear fallback
No retrieval layer	The 30 test inputs contain the routing signal	A future policy question would change the decision

What would change our mind: a ticket whose correct route depends on an internal policy, catalogue or document absent from the input.

The boundary checks the model before work moves on

Every model call passes through one module. The model returns JSON, then code validates both the field shape and rules that JSON Schema alone cannot express.

1	2	3	4	5
Ticket	Prompt	JSON record	Pydantic rules	Route or failure

- The schema restricts labels and confidence to the expected shape.
- A low-confidence or other label must set needs_human_review to true.
- Validation errors can trigger at most two correction prompts.
- The stop reason is checked before the text is trusted.
- The ticket is untrusted data, not a new instruction to the system.

Few-shot was strongest in the local comparison

Run date: 9 November 2026 | Ollama local | llama3.2:3b | temperature 0 | seed 42 | test file: evals/test_cases.csv | 30 cases

Strategy	Correct	Accuracy	Median	P90	Input	Output
Zero-shot	22 / 30	73.3%	4.33 s	6.92 s	6,911	680
Few-shot	25 / 30	83.3%	2.12 s	3.18 s	10,091	681
Chain-of-thought	22 / 30	73.3%	8.40 s	11.02 s	8,141	2,922

Few-shot is the provisional local recommendation. Local external cost was EUR 0 because the model ran on Ollama; this is not a live provider cost estimate.

The evidence has a clear limit

The run records five wrong few-shot cases and eight wrong cases for each of zero-shot and chain-of-thought. The final deck must show one real row from the result CSV: ticket text, predicted label, checked label and the reason for the error. We do not invent that example.

What passed	57 tests passed across the repository; the guardrail suite passed 27 tests.
What was skipped	3 live-model guardrail tests were skipped because no valid API key was available.
What remains unknown	Provider-specific accuracy, API latency, token pricing and judge calibration.

The defensible claim is limited: the local surrogate shows a comparison of prompt strategies, not production readiness.

Deploy the smallest measured improvement, then re-measure

For the local experiment, choose few-shot: 83.3% accuracy, 2.12 second median latency and 3.18 second P90 latency. Keep the review flag and bounded validation path around it.

- Repeat the 30-case comparison with an approved provider and a valid personal key.
- Have another member verify every expected label and review every wrong case.
- Run judge calibration and the swapped-order check required by the evaluation brief.
- Change the recommendation if the provider run reverses the accuracy or latency result, or if the error pattern creates operational risk.

Closing line: we built a bounded routing component, measured its local behaviour, and made the uncertainty visible instead of presenting a surrogate result as production evidence.

Before submission: replace the row-level failure placeholder on slide 6 with an actual case from runs/results_few_shot.csv.